

Lab Task 02

Title: Effect of Image Filtering on Skin-Lesion Classification

Objective

To investigate how different spatial-domain image-processing filters affect the performance of pretrained deep-learning models for skin-lesion classification.

Dataset

Use the **HAM10000 skin-lesion dataset**. Students may use the complete dataset or an instructor-approved subset. Clearly report the selected classes and number of images.

Task

Based on the research article studied in **Lab Activity 1 (Task 01)**, perform the following:

1. Use the **best three pretrained models identified from Lab Activity 1** for the classification experiments:
 - **Model 1:** EfficientNet-B0
 - **Model 2:** ResNet18
 - **Model 3:** ResNet50
2. Establish a **baseline experiment** using the original, unfiltered images.
3. Apply the following image-processing filters:
 - Average/Mean filter
 - Gaussian filter
 - Median filter
 - Sharpening filter
 - Sobel edge filter
4. For each of the three pretrained models, compare classification performance:
 - Without filtering
 - With each selected filter
5. Keep the **same dataset split, preprocessing, training settings, and evaluation procedure** across experiments to ensure a fair comparison.
6. Compare the effect of filtering on the three pretrained models and explain whether filtering consistently affects the models in the same way.

Required Experimental Results

Prepare a comparison table such as:

Model	Filter	Accuracy	Precision	Recall	F1-score	Macro-F1	AUC
EfficientNet-B0	No Filter	91.0	91.55	91.0	90.97	90.97	99.41
EfficientNet-B0	Average	81.0	81.17	81.0	80.35	80.35	97.79

EfficientNet-B0	Gaussian	88.0	88.72	88.0	88.01	88.01	98.46
EfficientNet-B0	Median	84.0	83.82	84.0	83.72	83.72	98.28
EfficientNet-B0	Sharpening	91.0	91.81	91.0	90.86	90.86	99.56
EfficientNet-B0	Sobel	80.0	80.79	80.0	79.81	79.81	95.95
ResNet18	No Filter	83.0	84.12	83.0	83.25	83.25	96.91
ResNet18	Average	89.0	89.71	89.0	89.11	89.11	97.48
ResNet18	Gaussian	89.0	88.95	89.0	88.70	88.70	96.60
ResNet18	Median	90.0	90.71	90.0	89.88	89.88	97.70
ResNet18	Sharpening	96.0	96.07	96.0	95.94	95.94	99.70
ResNet18	Sobel	78.0	80.31	78.0	78.19	78.19	95.65
ResNet50	No Filter	91.0	91.50	91.0	90.92	90.92	98.48
ResNet50	Average	86.0	88.03	86.0	84.64	84.64	95.90
ResNet50	Gaussian	86.0	87.48	86.0	86.08	86.08	97.84
ResNet50	Median	87.0	87.93	87.0	87.34	87.34	98.30
ResNet50	Sharpening	92.0	92.19	92.0	92.07	92.07	99.04
ResNet50	Sobel	80.0	83.23	80.0	80.62	80.62	95.78

Additional Analysis

For the experiments, include:

- HAM10000 class distribution
- Original and filtered image examples
- Confusion matrices
- Training/validation accuracy curves
- Training/validation loss curves
- Per-class precision, recall and F1-score
- Macro-F1
- Balanced accuracy
- AUC/ROC where applicable

Questions to Answer

1. Which three pretrained models performed best in Lab Activity 1?
2. How does filtering affect each of the three models?
3. Which filter produces the greatest change compared with the unfiltered baseline?
4. Does the effect of a filter remain consistent across all three models?
5. Does filtering improve or decrease macro-F1 and balanced accuracy?
6. Which lesion classes are most affected by filtering?
7. Why might smoothing remove useful lesion texture or morphological information?
8. Why might sharpening or edge detection help or hurt classification?
9. What is the difference between convolution and correlation?
10. Based on your results, explain the relationship between classical image processing and deep-learning-based feature extraction.

Code Submission

Students must:

1. Develop the complete experimental code in **Google Colab/Jupyter Notebook**.
2. Organize the code clearly into separate sections for:
 - Dataset preparation
 - Model loading
 - Baseline experiment
 - Image filtering
 - Training
 - Evaluation
 - Visualization
 - Comparative analysis
3. **Upload/push the complete code to the GitHub repository under Lab 02.**
4. The GitHub Lab 02 folder should contain the notebook/code and a brief README explaining how to run the experiments.